Module Handbook (excerpt) - M.Sc. Data & Knowledge Engineering

Programme structure: 120 CP total = 90 CP coursework + 30 CP master thesis. Recommended workload: ~30 CP per semester.

Thematic Area CP Requirements

Each student must earn credits within the following thematic areas:

Thematic Area	Min CP	Max CP
Fundamentals of Data Science	12	18
Learning Methods & Models	18	36
Data Processing for Data Science	18	30
Applied Data Science	18	24

Selectable Modules

Module	CP	Offered	Key skills taught	Prerequisites	Thematic Area
Advanced Databases	6	Winter	SQL, query optimization, indexing, transactions	none	Fundamentals of Science
Machine Learning Foundations	6	Winter	Supervised learning, model evaluation, scikit-learn	none	Fundamentals of Science
Distributed Systems	6	Winter	Replication, consensus, fault tolerance	none	Fundamentals of Science
Cloud Computing	6	Winter	IaaS/PaaS, virtualization, containers, Docker	none	Data Processing for Science
Data Mining I	6	Summer	Clustering, association rules, anomaly detection	none	Learning Methods Models
Information Retrieval	6	Winter	Search engines, ranking, text indexing	none	Learning Methods Models
Advanced Machine Learning	6	Summer	Deep learning, ensembles, hyperparameter tuning	Machine Learning Foundations	Learning Methods Models
Database Systems Implementation	6	Winter	Storage engines, query execution, buffer management	Advanced Databases	Learning Methods Models
Big Data Engineering	6	Summer	Spark, ETL pipeline design, data lakes, Airflow	Advanced Databases	Data Processing for Science
Stream Processing	6	Winter	Kafka, Flink, real-time pipelines, windowing	Big Data Engineering	Data Processing for Science
Data Warehouse Technologies	6	Winter	Dimensional modeling, OLAP, ELT processes	Advanced Databases	Data Processing for Science
MLOps in Practice	6	Summer	CI/CD for ML, model serving, monitoring, MLflow	Machine Learning Foundations	Data Processing for Science
Software Engineering for Data Science	6	Summer	Testing, CI/CD, clean code, agile practices	none	Applied Data Science
Data Visualization	3	Summer	Dashboards, visual analytics, data storytelling	none	Applied Data Science
Scientific Team Project	6	Winter & Summer	Team-based applied research project	none; at most twice	Applied Data Science
Seminar Data Engineering	3	Winter & Summer	Literature research, scientific writing, presentation	none; at most twice	Applied Data Science
Master Thesis	30	Winter & Summer	Independent scientific research	60 CP completed	Master Thesis

Notes

Modules marked 'Winter & Summer' run every semester. The Seminar and the Scientific Team Project may each be taken at most twice. The Master Thesis requires at least 60 CP of completed coursework.